PROJECT STATUS REPORT

Pediatric Patient Management and Clinical Database System

Evaluation 1 completion summary, verification record, and remaining work plan

CURRENT MILESTONE	RUNNING STATUS	DATA
Evaluation 1 Core relational system	**Website and API active** localhost development setup	**Synthetic only** 4 patients, 3 visits seeded

Overall position

The Evaluation 1 implementation is substantially complete and runnable. The core MySQL database, seed data, API, and patient-facing demonstration interface are working together. The remaining Evaluation 1 work is mainly evidence preparation and presentation polish, not major system construction.

IMPLEMENTATION	DOCUMENTATION	EVALUATION READINESS
Complete Core system built	**Complete** Design and demo guides	**Nearly ready** Needs final evidence capture

1. What has been completed

The items below are present in the repository and form the active Evaluation 1 demonstration.

DONE	**Database foundation** 26 core relational tables cover users and roles, departments and doctors, patients and guardians, appointments, visits, medical records, diagnoses, treatments, pediatric vital and growth records, allergies, immunizations, medications, prescriptions, laboratory work, follow-ups, and audit history.
DONE	**Relational integrity** Primary keys, foreign keys, unique rules, NOT NULL rules, CHECK constraints, junction tables, cascade behavior, and clinically relevant date/range validation are implemented.
DONE	**Indexes and query design** Composite indexes support patient timelines, doctor schedules, follow-up queues, prescribing history, lab workflows, and common filtering and joins.
DONE	**Reusable database logic** Three reporting views, one stored procedure, and audit triggers are included. A multi-step appointment and audit transaction demonstrates atomic consistency.
DONE	**Synthetic seed dataset** The database loads four fictional pediatric patients, guardians, doctors, appointments, three completed visits, diagnoses, observations, prescriptions, laboratory results, and follow-ups.
DONE	**SQL demonstration pack** Twenty meaningful SQL examples cover joins, grouping, subqueries, a window function, reporting, longitudinal history, doctor workload, medication use, age bands, and audit history.
DONE	**Application layer** A React interface and Express API provide a dashboard, patient search, patient registration, and longitudinal patient details. The application is linked to the MySQL database.
DONE	**Project documentation** README setup instructions, ER model, architecture explanation, normalization notes, Evaluation 1 demonstration sequence, and performance-query examples are available.

Verification completed

- React production build completed successfully.
- Server JavaScript syntax checks passed.
- An isolated MySQL instance was initialized on port 3307.
- Schema, views/routines, and sample data loaded successfully.
- Database check returned 4 patients and 3 visits.
- The running website was visually verified with dashboard totals, appointments, and recent visits.

2. What is left for Evaluation 1

These tasks complete the academic submission and live evaluation evidence. They are smaller than the implementation already finished.

HIGH	**Capture EXPLAIN evidence** Run the selected queries in MySQL Workbench with EXPLAIN or EXPLAIN ANALYZE. Save screenshots showing the chosen indexes, estimated/actual rows, and execution time.
HIGH	**Prepare the final ER diagram export** Export the Mermaid ER model or reverse-engineer the schema in MySQL Workbench and produce a clean image suitable for the evaluation presentation.
HIGH	**Record constraint demonstrations** Prepare two short demonstrations, such as a duplicate MRN and an invalid visit time, and show MySQL rejecting both operations.
MEDIUM	**Create presentation slides** Convert the existing design and Evaluation 1 demonstration guide into a concise slide deck with schema, normalization, query, transaction, index, and application evidence.
MEDIUM	**Add automated API tests** Add repeatable tests for patient search, patient registration transaction, error responses, and dashboard totals.
MEDIUM	**Finalize submission packaging** Choose the required submission format, include setup instructions and screenshots, and confirm that the evaluator can initialize the database from a clean machine.
OPTIONAL	**Add authentication demonstration** The schema includes users, roles, and permissions, but login and authorization middleware are intentionally outside the current academic prototype. Add only if required by the evaluator.

Recommended completion order

1. Capture Workbench ER and EXPLAIN screenshots. 2. Rehearse constraint and transaction demonstrations. 3. Build slides from the existing documentation. 4. Run a clean-machine setup test. 5. Add automated tests or authentication only if required.

3. Later milestone: Evaluation 2

Evaluation 2 is not part of the current milestone. A technical scaffold exists in the repository, but it should be presented only after Evaluation 1 has been accepted.

Scaffold already prepared

- Seven metadata-driven tables for forms, sections, fields, options, placement, typed values, and assignments.
- Backend routes for reading forms, adding a field, reordering fields, and saving clinical responses.
- A React form-builder/renderer page exists in source but is intentionally not linked in the active Evaluation 1 navigation.
- A configurable synthetic scale-data generator and advanced performance-query script are included as a starting point.

Still required for Evaluation 2

- Enable and fully test the form-builder user journey in the active application.
- Add form version history, stronger validation enforcement, and authorization for configuration changes.
- Generate benchmark datasets at 10,000, 50,000, and 100,000 patients, then capture comparable measurements.
- Complete before/after index experiments and dynamic-field query optimization.
- Add backup/restore and migration demonstrations if the evaluator requests them.
- Perform complete end-to-end and load testing for the extended system.

4. Key project locations

Database schema	`database/migrations/001_schema.sql`
Views and routines	`database/migrations/002_views_routines.sql`
Synthetic seed data	`database/seeds/001_seed.sql`
Evaluation queries	`database/queries/evaluation_queries.sql`
Design and ER model	`docs/DESIGN.md`
Evaluation 1 demo guide	`docs/EVALUATION_1.md`
Application	`client/ and server/`

Conclusion

Evaluation 1 has a functioning technical implementation and is ready to move into evidence capture, presentation preparation, and final rehearsal. Evaluation 2 should remain a separate later milestone.